

Journal #86 - November 2017

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Cosmotechnics as Cosmopolitics



The main hall of the Chinese Science and Technology Museum, Beijing, August 2010.

The end of unilateral globalization and the arrival of the Anthropocene force us to talk about cosmopolitics. These two factors correlate with one another and correspond to two different senses of the word “cosmopolitics”: cosmopolitics as a commercial regime, and cosmopolitics as a politics of nature.

First, we are witnessing the end of unilateral globalization. Until now, so-called globalization has been a largely one-sided process, entailing the universalization of particular epistemologies and the elevation, through techno-economic means, of a regional worldview to a putatively global metaphysics. We know that this unilateral globalization has reached its end because of how the 9/11 attacks were misread as an attack on the Occident by an Other. In fact, 9/11 was an “autoimmune” event, internal to the Atlantic bloc, wherein its own anti-communist cells, lingering after the Cold War, turned against their hosts.¹ Still, the spectacular image of the event provided a kind of Rorschach test, onto which the representatives of unilateral globalization could project their growing insecurities about being stranded between the old configuration and the new—exemplifying what Hegel called “the unhappy consciousness.”² This is clear in an article entitled “The Straussian Moment” by one of the leading financiers of American neoreaction, Peter Thiel:

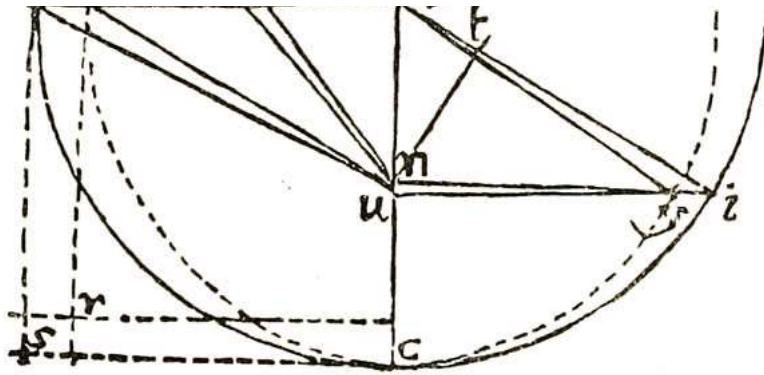


Diagram used by Johannes Kepler to establish his laws of planetary motion. Photo: Wikimedia Commons.

§1. Cosmopolitanism: Between Nature and Technology

The main difficulty of all cosmopolitics is the reconciliation between the universal and the particular. The universal tends to contemplate the particulars from above, as in the way that Kant regarded the French Revolution, like a spectator considering a violent piece of theater from the mezzanine. Universality is the view of a spectator, never that of an actor. Kant writes, in his “Idea for a Universal History with a Cosmopolitan Aim”:

There is no other way out for the philosopher—who, regarding human beings and their play in the large, cannot at all presuppose any rational aim of theirs—than to try whether he can discover an aim of nature in this nonsensical course of things human; from which aim a history in accordance with a determinate plan of nature might nevertheless be possible even of creatures who do not behave in accordance with their own plan ... [Nature] did produce a Kepler, who subjected the eccentric paths of the planets in an unexpected way to determinate laws, and a Newton, who explained these laws from a universal natural cause.⁷

Throughout his political writings, Kant maintains that this relation between nature and cosmopolitics is necessary.⁸ If Kant sees the republican constitution and perpetual peace as political forms that may be able to bring forward a universal history of the human species, it is because he understands that such progress is also a progress of reason, the telos of nature. This progress toward an end goal—namely, universal history and a “perfect state constitution”—is the “completion of a hidden plan of nature” (*Vollziehung eines verborgenen Plans der Natur*). What does it mean for nature to have a hidden plan? And why is the realization of cosmopolitics the teleology of nature?

Authors such as Hannah Arendt and Eckart Förster, among others, suggest that Kant’s political philosophy centers on his concept of nature.⁹ Arendt proposes a juxtaposition concerning Kant’s perpetual peace: on the one hand, *Besuchsrecht*, the right to visit foreign countries and the right to hospitality; and on the other, nature, “the great artist, as the eventual ‘guarantee of perpetual peace.’”¹⁰ If after the 1789 revolution Kant is even more consistent in his affirmation of cosmopolitics as the teleology of nature, it is because he has developed the concept of self-organization, which plays a central role in the second book of his *Critique of Judgment*, and which affirms the two important categories of relation, namely community (*Gemeinschaft*) and reciprocity (*Wechselwirkung*).

Consider Kant’s example of the tree from §64 of the *Critique of Judgment*. First, the tree reproduces itself according to its genus, meaning that it reproduces another tree. Second, the tree produces itself as an individual; it absorbs energy from the environment and turns it into nutrients that sustain its life. Third, different parts of the tree establish reciprocal relations with one another and thus constitute the whole; as Kant writes, the “preservation of one part is reciprocally dependent on the preservation of the other parts.”¹¹ In such a totality, a part is always constrained by the whole, and this is true of Kant’s understanding of cosmopolitical wholeness as well: “All states ... are in danger of acting injuriously upon one another.”¹² Nature is not something that can be judged from a particular point of view, just as the French Revolution cannot be judged according to its actors. Rather, nature can only be comprehended as a complex whole, and the human species, as one part of it, will ultimately progress towards a universal history that coincides with the teleology of nature.¹³

Here we only want to show that as Kant develops his thinking towards universalism, his conceptualization of the relation between cosmopolitics and the purposiveness of nature is situated within a peculiar moment in history: the simultaneous enchantment and disenchantment of nature. On the one hand, Kant recognizes the importance of the concept of the organic for philosophy; discoveries in the natural sciences allowed him to connect the cosmos to the moral, as indicated by his famous analogy near the end of *Critique of Practical Reason*: “Two things fill the mind with ever new and increasing wonder and awe, the more often and constantly reflection concerns itself with them: the starry heavens above me and the moral law within me.”¹⁴ Howard Caygill makes an even stronger claim, arguing that this analogy points to a “Kantian physiology of the soul and the cosmos” that unites the “within me” (freedom) and the “above me.”¹⁵ On the other hand, as we saw in Kant’s citation of Kepler and Newton in “Idea for a Universal History with a Cosmopolitan Aim,” the affirmation of “universal history” and advancements in science and technology led in the eighteenth century to what Rémi Brague calls the “death of the cosmos”:

The new astronomy, following Copernicus and his successors, had consequences for the modern view of the world ... Ancient and medieval thinkers presented a synchronic schema of the structure of the physical world, which erased the traces of its own genesis; the Moderns, on the other hand, remembered the past and in addition provided a diachronic view of astronomy —as if the evolution of ideas about the cosmos was even more important than the truth about it ... Can we still speak of cosmology? It seems that the West ceased to have a cosmology with the end of the world of Aristotle and Ptolemy, an end due to Copernicus, Galileo, and Newton. The “world” then no longer formed a whole.¹⁶

New discoveries in the natural sciences thanks to the invention of the telescope and the microscope exposed human beings to magnitudes they could not previously comprehend, leading us to a new relation with the “entire span of nature” (*in dem ganzen Umfang der Natur*).¹⁷ The Kantian scholar Diane Morgan suggests that through the “worlds beyond worlds” revealed by technology, nature ceases to be anthropomorphic, for the relation between humans and nature is thus reversed, with humans now standing before the “unsurveyable magnitude” (*Unabsehlich-Groß*) of the universe.¹⁸ However, as we indicated above, there is a double moment that deserves our attention: both the enchantment *and* disenchantment of nature via the natural sciences, leading to a total secularization of the cosmos.

In addition to the revelation of nature and its teleology through technical instruments, technology also plays a decisive role in Kant’s political philosophy, when he asserts that communication is the condition of the realization of the organicist whole. Arendt made explicit the role of the *sensus communis* in Kant’s philosophy, as both the question of community and consensus.¹⁹ But such a *sensus communis* is achieved only through particular technologies, and it is on this ground that we should problematize any naive discourse on the common as something already given or preceding technology. The age of Enlightenment, as noted by Arendt (as well as Bernard Stiegler), is the age of “the public use of one’s reason,” and this exercise of reason is expressed in the freedom of speaking and publishing, which necessarily involves the technology of printing. On an international level, in “Toward Perpetual Peace: A Philosophical Sketch” Kant writes that “it was trade that first brought them into peaceful relations with one another and thereby into relationships based on mutual consent, community, and peaceful interactions even with remote peoples,” later adding, “it is the spirit of trade, which cannot coexist with war, which will, sooner or later, take hold of every people.”²⁰





Gisela Motta and Leandro Lima, *Xaipiri*, 2012.

§2. “Ontological Turn” as *Cosmopolitics*

This reiteration of Kantian cosmopolitanism is an attempt to demonstrate the role of nature in Kant’s political philosophy. Kant somehow assumes *one single nature*, which reason compels us to recognize as rational; the rationality corresponds to the organicist teleological universality ostensibly realized in the constitution of both morality and the state. This enchantment of nature is accompanied by a disenchantment of nature, driven by the mechanization enforced by the Industrial Revolution. Brague’s “death of the cosmos” brought about by European modernity and its globalization of modern technology necessarily forms one of the conditions for us to reflect on cosmopolitics today, insofar as it illustrates the inefficacy of a biological metaphor for cosmopolitanism. If we start with Kant rather than with more recent discussions on cosmopolitanism—such as Martha Nussbaum’s rootless cosmopolitanism, Habermas’s constitutional patriotism, or Anthony Appiah’s cosmopolitan patriotism²¹—it is because we want to reconsider cosmopolitanism by examining its relation to nature and technology. In fact, Appiah’s rooted cosmopolitanism is relevant to our discussion below. He holds the view that cosmopolitanism denies the importance of affiliations and particular loyalties; this means that it is necessary to consider cosmopolitics from the point of view of locality. This crucial point is the reason I would like to engage with the idea of “multi-naturalism” recently proposed by anthropologists associated with attempts to present a new way of thinking cosmopolitanism.

The “ontological turn” in anthropology is a movement associated with anthropologists such as Philippe Descola, Eduardo Viveiros de Castro, Bruno Latour, and Tim Ingold, and earlier, Roy Wagner and Marilyn Strathern, among others.²² This ontological turn is an explicit response to the crisis of modernity that expresses itself largely in terms of ecological crisis, which is now closely associated with the Anthropocene. The ontological-turn movement is an effort to take seriously different ontologies in different cultures (we have to bear in mind that knowing there are different ontologies and taking them seriously are two different things). Descola has convincingly outlined four major ontologies, namely naturalism, animism, totemism, and analogism.²³ The modern is characterized by what he calls “naturalism,” meaning an opposition between culture and nature, and the former’s mastery over the latter. Descola suggests that we must go beyond such an opposition and recognize that nature is no longer opposed or inferior to culture. Rather, in the different ontologies, we can see the different roles that nature plays; for example, in animism the role of nature is based on the continuity of spirituality, despite the discontinuity of physicality.

In *Beyond Culture and Nature*, Descola has proposed an ontological pluralism that is irreducible to social constructivism. He suggests that recognizing these ontological differences can serve as an antidote to the dominance of naturalism since the advent of European modernity. But does this focus on nature (or the cosmos, we might say) in the interest of opposing European naturalism actually revive the enchantment of nature, this time in the name of indigenous knowledge? This seems to be a hidden problem with the ontological-turn movement: many anthropologists associated with the ontological turn have focused on the question of nature and the politics of the nonhuman (largely animals, plants, minerals, spirits, and the dead). This is evident when we recall that Descola proposes to call his discipline an “anthropology of nature.” Furthermore, this tendency also suggests that the question of technics is not sufficiently addressed in the ontological-turn movement. For example, Descola talks often of practice, which may indicate his (laudable) desire to avoid an opposition between nature and technics; but by doing so, he also obscures the question of technology. Descola shows that analogism, rather than naturalism, was a significant presence in Europe during the Renaissance; if this is the case, the “turn” that took place during European modernity seems to have resulted in a completely different ontology and epistemology. If naturalism has succeeded in dominating modern thought, it is because such a peculiar cosmological imagination is compatible with its *techno-logical* development: nature should be mastered for the good of man, and it *can* indeed be mastered according to the laws of nature. Or put another way: nature is regarded as the source of contingency due to its “weakness of concept,” and therefore it has to be overcome by logic.

These oppositions between nature and technics, mythology and reason, give rise to various illusions that belong to one of two extremes. On the one hand, there are rationalists or “progressivists” who hysterically struggle to maintain their monotheism after having murdered god, wishfully believing that the world process will stamp out differences and diversities and lead to a “theodicy.” On the other hand, there are left intellectuals who feel the need to extol indigenous ontology or biology as a way out of modernity. A French revolutionary thinker recently described this situation thus:

A funny thing to see these days is how all these absurd modern leftists, all unable to *see* anything, all lost in themselves, all feeling so bad, all desperately trying to exist and to find their existence in the eyes of the Other—how all these people are jumping on the “savage,” the “indigenous,” the “traditional” in order to escape and not face themselves. I am not speaking of being critical towards one’s “whiteness,” towards one’s “modernism.” I am talking of the ability to peer inside [*transpercer*] oneself.

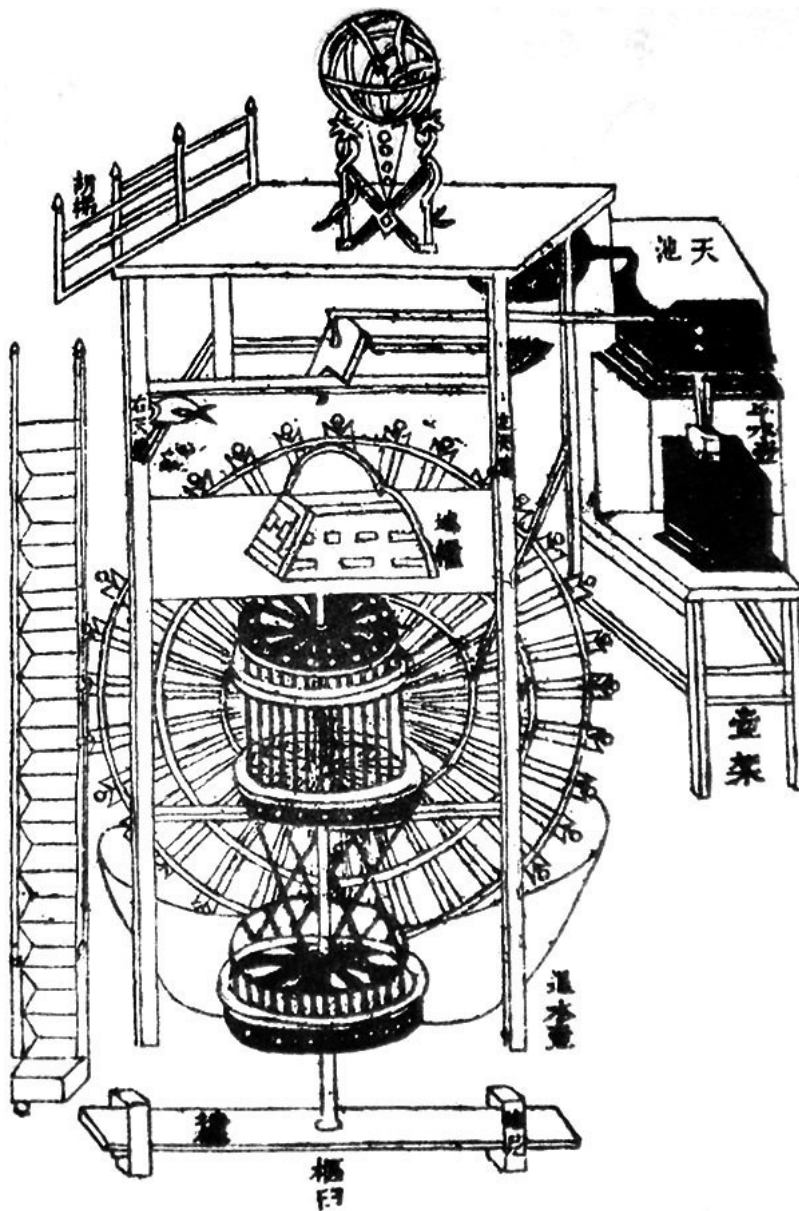
My refusal of the above two extremes does not come out of any postcolonial “political correctness,” but rather out of an attempt to go beyond postcolonialism’s critique. (Indeed, I have elsewhere reproached postcolonialism for its failure to tackle the question of technology.²⁴) I hold the thesis that an ontological pluralism can only be realized by reflecting on the question of technology and a politics of technology. Kant was aware of the importance of technology in his comment on trading as communication; however, he didn’t pay much attention to the *technological difference* that finally led to planetary modernization, and now planetary computation, since what was at stake for him was the question of the whole that absorbs all differences. Kant criticized the impolite guests, the greedy colonizers who brought with them “oppression of the native inhabitants, the incitement of the different states involved to expansive wars, famine, unrest, faithlessness, and the whole litany of evils that weigh upon the human species.”²⁵ Commenting on the defense strategies of China and Japan, Kant said that both countries have

wisely, limited such interaction. Whereas the former has allowed contact with, but not entrance to its territories, the latter has allowed this contact to only one European people, the Dutch, yet while doing so it excludes them, as if they were prisoners, from associating with the native inhabitants.²⁶

When Kant wrote this in 1795, it was too early for him to anticipate the modernization and colonization that would take place in Japan and China. If this phase of globalization was able to take place, it was because of the technological advancement of the West, which allowed it to defeat the Japanese, the Chinese, and other Asian civilizations. Nature, the guarantee of perpetual peace, didn’t really lead us to perpetual peace but rather to wars and more wars. To appeal for a cosmopolitanism today, I think we must reread Kant’s cosmopolitanism according to the process of modernization and revisit the question of nature and technology anew. The arrival of

modern technology in non-European countries in recent centuries has created a transformation unthinkable to European observers. The restoration of “indigenous natures” itself has to first be questioned, not because it doesn’t exist but because it is situated in a new epoch and is transformed to the extent that there is hardly any way to go back and restore it.²⁷

Let’s review what has been said above regarding the ontological turn. Central to the anthropologists’ concept of “nature” and “ontology” is cosmology, since such “nature” is defined according to different “ecologies of relations” in which we observe different constellations of relations, e.g., the parental relation between females and vegetables, or brotherhood between hunters and animals. These multi-ontologies are expressed as multi-natures; for example, Descola’s four above-named ontologies correspond to different cosmological views. I believe that it is very difficult, if not impossible, to overcome modernity without directly confronting the question of technology, which has become increasingly urgent after the end of unilateral globalization. Therefore, it is necessary to reformulate the question of cosmopolitics in relation to cosmotechnics.



A diagram of Su Song's (1020–1101) clock tower. The original design included an armillary sphere, a waterwheel, an escapement mechanism, and a chain drive. Photo: Wikimedia Commons.

§3. Cosmotechnics as Cosmopolitics

I propose to go beyond the notion of cosmology; instead, it would be more productive to address what I call cosmotechnics. Let me give you a preliminary definition of cosmotechnics: it is the unification of the cosmos and

the moral through technical activities, whether craft-making or art-making. There hasn't been one or two technics, but many cosmotechnics. What kind of morality, which and whose cosmos, and how to unite them vary from one culture to another according to different dynamics. I am convinced that in order to confront the crisis that is before us—namely, the Anthropocene, or the intrusion of Gaia (Latour and Stengers), or the “entropocene” (Stiegler), all presented as the inevitable future of humanity—it is necessary to reopen the question of technology, in order to envisage the bifurcation of technological futures by conceiving different cosmotechnics. I tried to demonstrate such a possibility in my recent book *The Question Concerning Technology in China: An Essay in Cosmotechnics*. As one can gather from the title, it is an attempt to respond to Heidegger's famous 1949 lecture “The Question Concerning Technology.” I propose that in order to rethink the project of overcoming modernity, we must undo and redo the translations of *technē*, *physis*, and *metaphysika* (not as merely independent concepts but also concepts within systems); only by recognizing this difference can we arrive at the possibility of a common task of philosophy.

Why, then, do I think it's necessary to turn to cosmotechnics? For a long time now we have operated with a very narrow—in fact, far too narrow—concept of technics. By following Heidegger's essay, we can distinguish two notions of technics. First, we have the Greek notion of *technē*, which Heidegger develops through his reading of the ancient Greeks, notably the Pre-Socratics—more precisely, the three “inceptual” (*anfängliche*) thinkers, Parmenides, Heraclitus, and Anaximander.²⁸ In the 1949 lecture, Heidegger proposes to distinguish the essence of Greek *technē* from modern technology (*moderne Technik*).

If the essence of *technē* is *poiesis*, or bringing forth (*Hervorbringen*), then modern technology, a product of European modernity, no longer possesses the same essence as *technē* but is rather an “enframing” (*Gestell*) apparatus, in the sense that all beings become standing reserves (*Bestand*) for it. Heidegger doesn't totalize these two essences of technics, but nor does he give space to other technics, as if there is only a single homogenous *Machenschaft* after the Greek *technē*, one that is calculable, international, even planetary. It is astonishing that in Heidegger's so-called *Black Notebooks* (*Schwarze Hefte*)—of which four volumes have been published so far—we find this note: “If communism in China should come to rule, one can assume that only in this way will China become ‘free’ for technology. What is this process?”²⁹ Heidegger hints at two things here: first, that technology is international (not universal); and second, that the Chinese were completely unable to resist technology after communism seized power in the country. This verdict anticipates technological globalization as a form of neocolonization that imposes its rationality through instrumentality, like what we observe in transhumanist, neoreactionary politics.

My effort to go beyond Heidegger's discourse on technology is largely based on two motivations: 1) a desire to respond to the ontological turn in anthropology, which aims to tackle the problem of modernity by proposing an ontological pluralism; and 2) a desire to update the insufficient discourse on technology that is largely associated with Heidegger's critique of technology. I have proposed that we reopen the question of technics, to show that one must consider technics as a variety of cosmotechnics instead of either *technē* or modern technology. In my book, I used China as a testing ground for my thesis and tried to reconstruct a lineage of technological thought in China. However, this task is not limited to China, since the central idea is that every non-European culture must systematize its own cosmotechnics and the history of such a cosmotechnics. Chinese cosmotechnical thought consists of a long history of intellectual discourse on the unity and relation between Qi and Dao. The unification of Qi and Dao is also the unification of the moral and the cosmic, since Chinese metaphysics is fundamentally a moral cosmology or a moral metaphysics, as the New Confucian philosopher Mou Zongsan has demonstrated. Mou suggests that if in Kant we find a metaphysics of the moral, it is at most a metaphysical exploration of the moral but not a moral metaphysics, since a moral metaphysics can only start with the moral. Mou's demarcation between Chinese and Western philosophy situates his conviction that Chinese philosophy recognizes and cultivates the intellectual intuition that Kant associated with knowing the noumenon, even as Kant dismissed the possibility that human beings could possess such an intuition. For Mou, the moral arises out of the experience of the infinity of the cosmos, which necessitates infinitization as the condition of possibility for Dasein's finitude.³⁰

Dao is not a thing. It is not a concept. It is not the *différance*. In the *Cixi* of *YiZhuan* (易傳·繫辭), Dao is simply said to be “above forms,” while Qi is what is “below forms.”³¹ We should notice here that *xin er shang xue* (the study of what is above forms) is the word used to translate “metaphysics” (one of the equivalences that must be

undone). Qi is something that takes space, as we can see from the character and also read in an etymological dictionary—it has four mouths or containers and in the middle there is a dog guarding the utensils. There are multiple meanings of Qi in different doctrines; for example, in classic Confucianism there is Li Qi (禮器), in which Qi is crucial for Li (a rite), which is not merely a ceremony but rather a search for unification between the heavens and the human. For our purposes, it will suffice to simply say that Dao belongs to the noumenon according to the Kantian distinction, while Qi belongs to the phenomenon. But it is possible to infinitize Qi so as to infinitize the self and enter into the noumenon—this is the question of art.

In order to better understand what I mean by this, we can refer here to the story of the butcher Pao Ding, as told in the *Zhuangzi*. However, we will have to remind ourselves that this is only an example from antiquity, and a much larger historical view is necessary to comprehend it.

Pao Ding is excellent at butchering cows. He claims that the key to being a good butcher doesn't lie in mastering certain skills, but rather in comprehending the Dao. Replying to a question from Duke Wen Huei about the Dao of butchering cows, Pao Ding points out that having a good knife is not necessarily enough; it is more important to understand the Dao in the cow, so that one does not use the blade to cut through the bones and tendons, but rather to pass alongside them in order to enter into the gaps between them. Here, the literal meaning of “Dao”—“way” or “path”—meshes with its metaphysical sense:

What I love is Dao, which is much more splendid than my skill. When I first began to carve a bullock, I saw nothing but the whole bullock. Three years later, I no longer saw the bullock as a whole but in parts. Now I work on it by intuition and do not ^[1]_{SEP} look at it with my eyes. My visual organs stop functioning while my intuition goes its own way. In accordance with the principle of heaven (nature), I cleave along the main seams and thrust the knife into the big cavities. Following the natural structure of the bullock, I never touch veins or tendons, much less the big bones!³²

Hence, Pao Ding concludes that a good butcher doesn't rely on the technical objects at his disposal, but rather on Dao, since Dao is more fundamental than Qi (the tool). Pao Ding adds that a good butcher has to change his knife once a year because he cuts through tendons, while a bad butcher has to change his knife every month because he cuts through bones. Pao Ding, on the other hand—an *excellent* butcher—has not changed his knife in nineteen years, and it looks as if it has just been sharpened with a whetstone. Whenever Pao Ding encounters any difficulty, he slows down the knife and gropes for the right place to move further.

Duke Wen Huei, who had posed the question, replies that “having heard from Pao Ding, now I know how to *live*”; and indeed, this story is included in a section titled “Master of Living.” It is thus the question of “living,” rather than that of technics, that is at the center of the story. If there is a concept of “technics” here, it is one that is detached from the technical object: although the technical object is not without importance, one cannot seek the perfection of technics through the perfection of a tool or a skill, since perfection can only be accomplished by Dao. Pao Ding's knife never cuts tendons or bones; instead, it seeks the void and enters it with ease. In so doing, the knife accomplishes the task of butchering the cow without endangering itself—i.e., without becoming blunt and needing to be replaced. It thus fully realizes itself as a knife.^[1]_{SEP}

What I have said above is not sufficient to be formulated into a program, since it is only an explanation for the motivation behind the much larger project that I tried to initiate in *The Question Concerning Technology in China*. Also, we must pay attention to the historical development of the relationship between Qi and Dao. Specifically, the search for unity between Qi and Dao has gone through different phases in Chinese history in response to historical crises (the decline of the Zhou Dynasty, the proliferation of Buddhism, modernization, etc.); it was widely discussed after the Opium Wars of the mid-nineteenth century, but such a unification was not resolved due to a very limited understanding of technology at the time and an eagerness to look for equivalences between China and the West. I have attempted to reread the history of Chinese philosophy not only as intellectual history, but also through the lens of the Qi-Dao episteme, with the aim of reconstructing a tradition of technological thought in China. As I have emphasized elsewhere, this question is by no means only a Chinese affair.³³ Rather, every culture must reflect on the question of cosmotechnics for a new cosmopolitics to come, since I believe that to overcome modernity without falling back into war and fascism, it is necessary to reappropriate modern technology through the renewed framework of a cosmotechnics consisting of different

epistemologies and epistemes. Therefore, my project is not one of substantializing tradition, as in the case of traditionalists like René Guénon or Aleksandr Dugin; it doesn't refuse modern technology, but rather looks into the possibility of different technological futures. The Anthropocene is the planetarization of standing reserves, and Heidegger's critique of technology is more significant today than ever before. The unilateral globalization that has come to an end is being succeeded by the competition of technological acceleration and the allures of war, technological singularity, and transhumanist (pipe) dreams. The Anthropocene is a global axis of time and synchronization that is sustained by this view of technological progress towards the singularity. To reopen the question of technology is to refuse this homogeneous technological future that is presented to us as the only option.

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The author would like to thank Pieter Lemmens and Kirill Chepurin for their comments on a draft of this essay.

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